

**SET 1 – Product Inventory Management****1. Bar Chart****AIM**

To create a bar chart to visualize the quantity available for each product.

**ALGORITHM**

1. Create vectors for product names and quantities.
2. Use barplot() to represent the inventory quantity.
3. Label the x-axis and y-axis.
4. Add a suitable title.
5. Display the bar chart.

**PROGRAM**

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

```
quantity <- c(250, 175, 300, 200, 220)
```

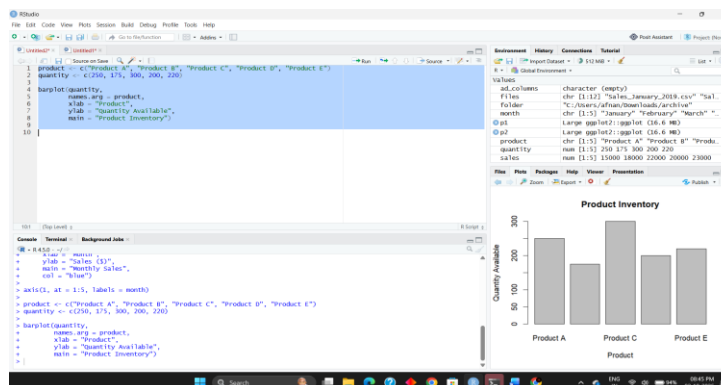
```
barplot(quantity,
```

```
names.arg = product,
```

```
xlab = "Product",
```

```
ylab = "Quantity Available",
```

```
main = "Product Inventory")
```

**OUTPUT:**

## RESULT

The bar chart successfully displays the quantity available for each product.

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## 2. Stacked Bar Chart

### AIM

To create a stacked bar chart showing product quantities according to product categories.

### ALGORITHM

1. Create vectors for product names, categories, and quantities.
2. Create a table using table().
3. Use barplot() to create a stacked bar chart.
4. Add labels, title, and legend.
5. Display the chart.

### PROGRAM

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

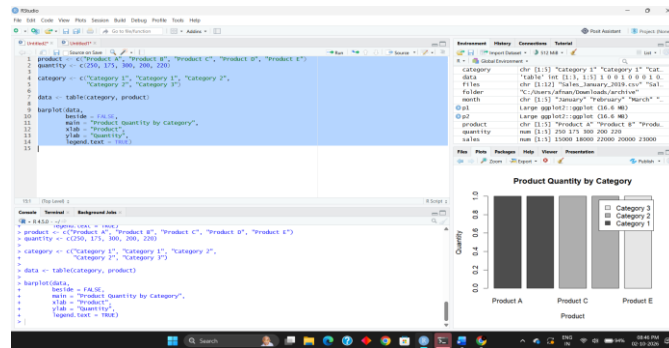
```
quantity <- c(250, 175, 300, 200, 220)
```

```
category <- c("Category 1", "Category 1", "Category 2",  
             "Category 2", "Category 3")
```

```
data <- table(category, product)
```

```
barplot(data,  
        beside = FALSE,  
        main = "Product Quantity by Category",  
        xlab = "Product",  
        ylab = "Quantity",  
        legend.text = TRUE)
```

## OUTPUT:



## RESULT

The stacked bar chart successfully represents product quantities by category.

## 3. Scatter Plot

### AIM

To explore the relationship between product price and quantity available using a scatter plot.

### ALGORITHM

1. Create vectors for product price and quantity.
2. Use `plot()` to create the scatter plot.
3. Label both axes.
4. Add a suitable title.
5. Display the plot and observe the relationship.

### PROGRAM

```
price <- c(100, 150, 120, 180, 130)
```

```
quantity <- c(250, 175, 300, 200, 220)
```

```
plot(price, quantity,
```

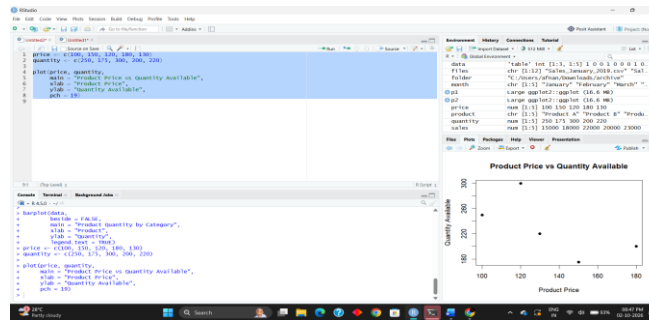
```
      main = "Product Price vs Quantity Available",
```

```
      xlab = "Product Price",
```

```
      ylab = "Quantity Available",
```

```
      pch = 19)
```

## OUTPUT:



## RESULT

The scatter plot successfully shows the relationship between product price and quantity available.

**Note:** The question does not provide product prices, so sample price values are used.

## SET 2 – Website Analytics

### 1. Line Chart

#### AIM

To visualize the trend in daily website page views using a line chart.

#### ALGORITHM

1. Create vectors for dates and page views.
2. Convert the dates into date format.
3. Use plot() with line type.
4. Label the axes and title.
5. Display the line chart.

#### PROGRAM

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",  
                  "2023-01-04", "2023-01-05"))
```

```
page_views <- c(1500, 1600, 1400, 1650, 1800)
```

```
plot(date, page_views,
```

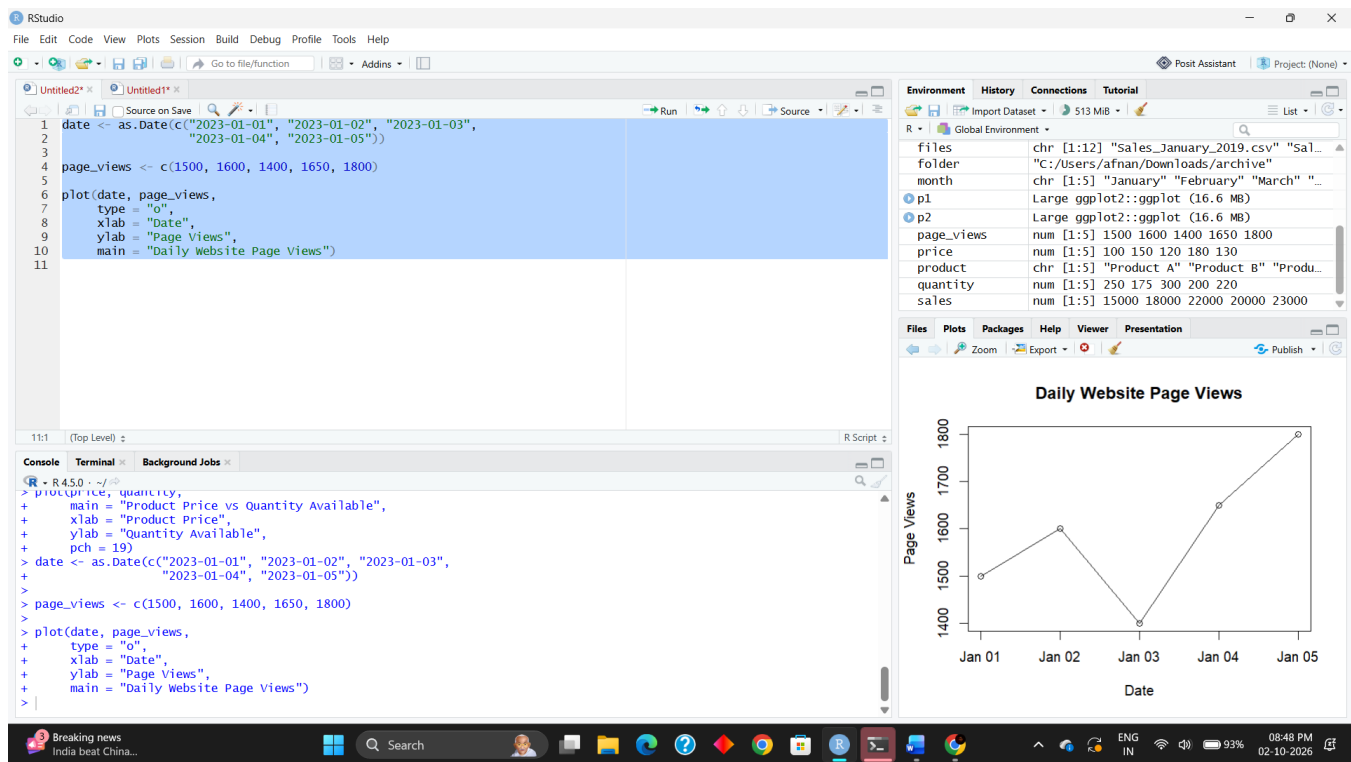
```
      type = "o",
```

```
      xlab = "Date",
```

```
ylab = "Page Views",

main = "Daily Website Page Views")
```

## OUTPUT:



## RESULT

The line chart successfully displays the daily trend in website page views.

## 2. Bar Chart

### AIM

To create a bar chart showing the click-through rate for each day.

### ALGORITHM

1. Create vectors for dates and click-through rates.
2. Use barplot() to create the chart.
3. Label the axes.
4. Add a suitable title.
5. Display the bar chart.

## PROGRAM

```
date <- c("Jan 1", "Jan 2", "Jan 3", "Jan 4", "Jan 5")
```

```
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)
```

```
barplot(ctr,
```

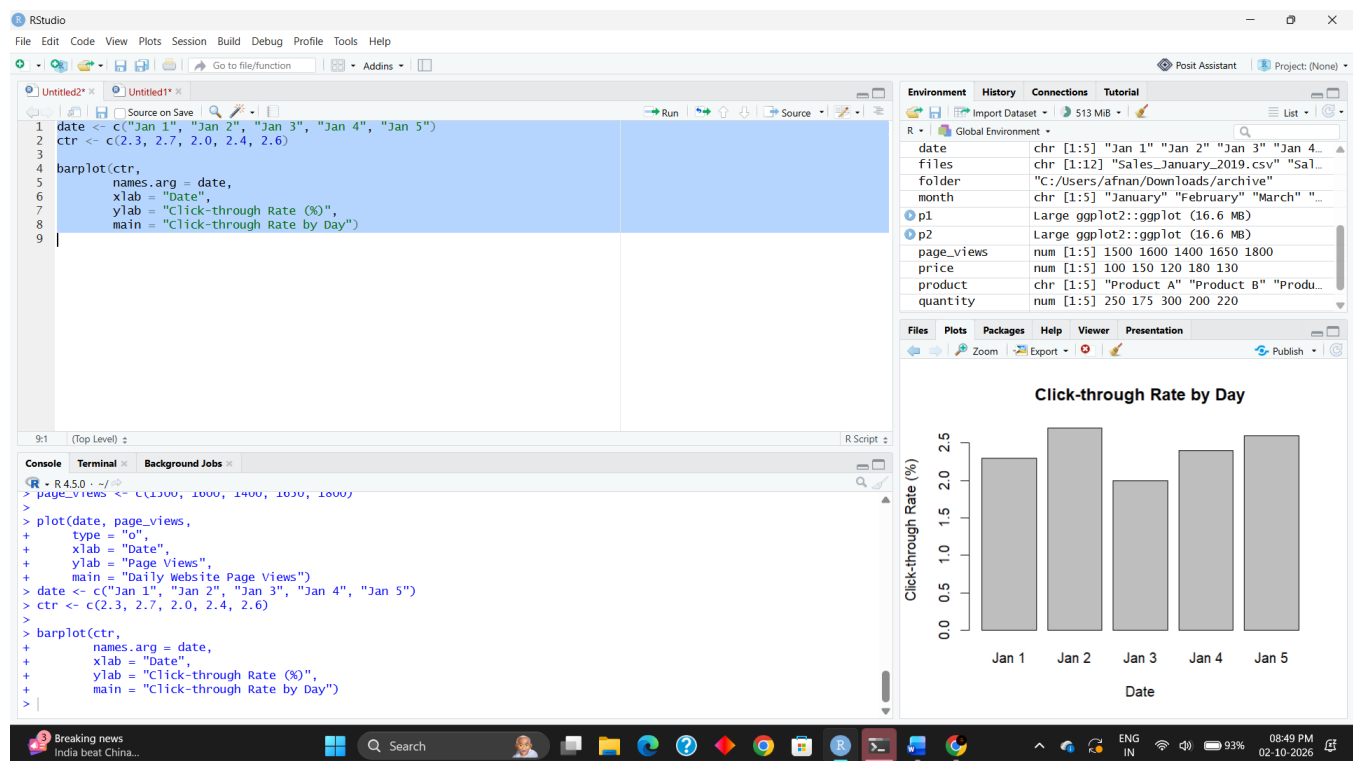
```
  names.arg = date,
```

```
  xlab = "Date",
```

```
  ylab = "Click-through Rate (%)",
```

```
  main = "Click-through Rate by Day")
```

## OUTPUT:



## RESULT

The bar chart successfully displays the click-through rate for each day.

## 3. Stacked Area Chart

### AIM

To display the distribution of website user interactions such as likes, shares, and comments using a stacked area chart.

## ALGORITHM

1. Create vectors for dates, likes, shares, and comments.
2. Calculate cumulative interaction values.
3. Use `polygon()` to create stacked areas.
4. Add labels and legend.
5. Display the chart.

## PROGRAM

```
date <- 1:5
```

```
likes <- c(100, 120, 90, 130, 150)
```

```
shares <- c(40, 50, 35, 55, 60)
```

```
comments <- c(30, 35, 25, 40, 45)
```

```
total1 <- likes
```

```
total2 <- likes + shares
```

```
total3 <- likes + shares + comments
```

```
plot(date, total3,  
      type = "n",  
      xlab = "Day",  
      ylab = "User Interactions",  
      main = "Website User Interactions")
```

```
polygon(c(date, rev(date)),  
        c(total1, rep(0, 5)))
```

```
polygon(c(date, rev(date)),  
        c(total2, rev(total1)))
```

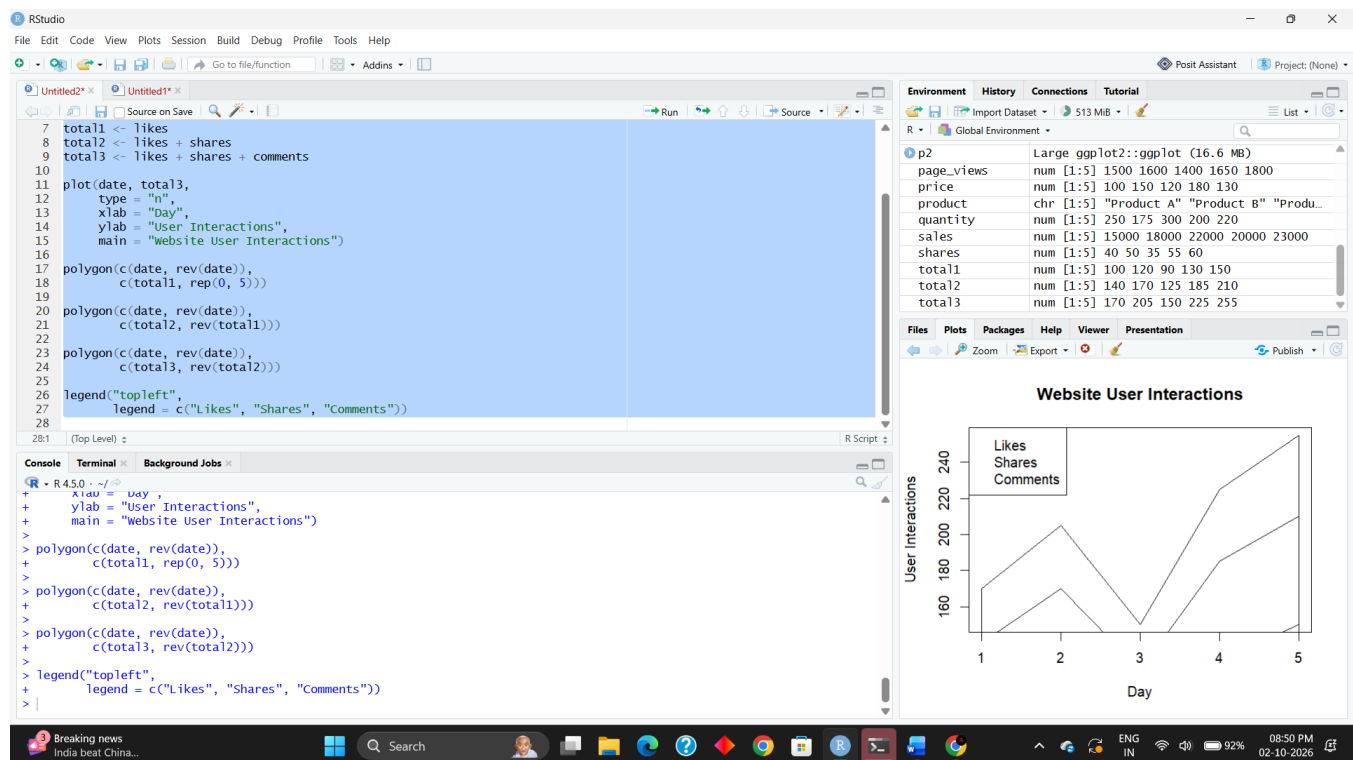
```

polygon(c(date, rev(date)),
        c(total3, rev(total2)))

legend("topleft",
       legend = c("Likes", "Shares", "Comments"))

```

## OUTPUT:



## RESULT

The stacked area chart successfully represents the distribution of likes, shares, and comments over time.

**Note:** Likes, shares, and comments were not included in the supplied dataset, so sample values are used.

## SET 3 – Product Sales Analysis

### 1. Grouped Bar Chart

#### AIM

To create a grouped bar chart to visualize the sales of each product during January, February, and March.

#### ALGORITHM

1. Create product and monthly sales vectors.
2. Combine the sales values into a matrix.

3. Use `barplot()` with `beside = TRUE`.
4. Add labels and legend.
5. Display the grouped bar chart.

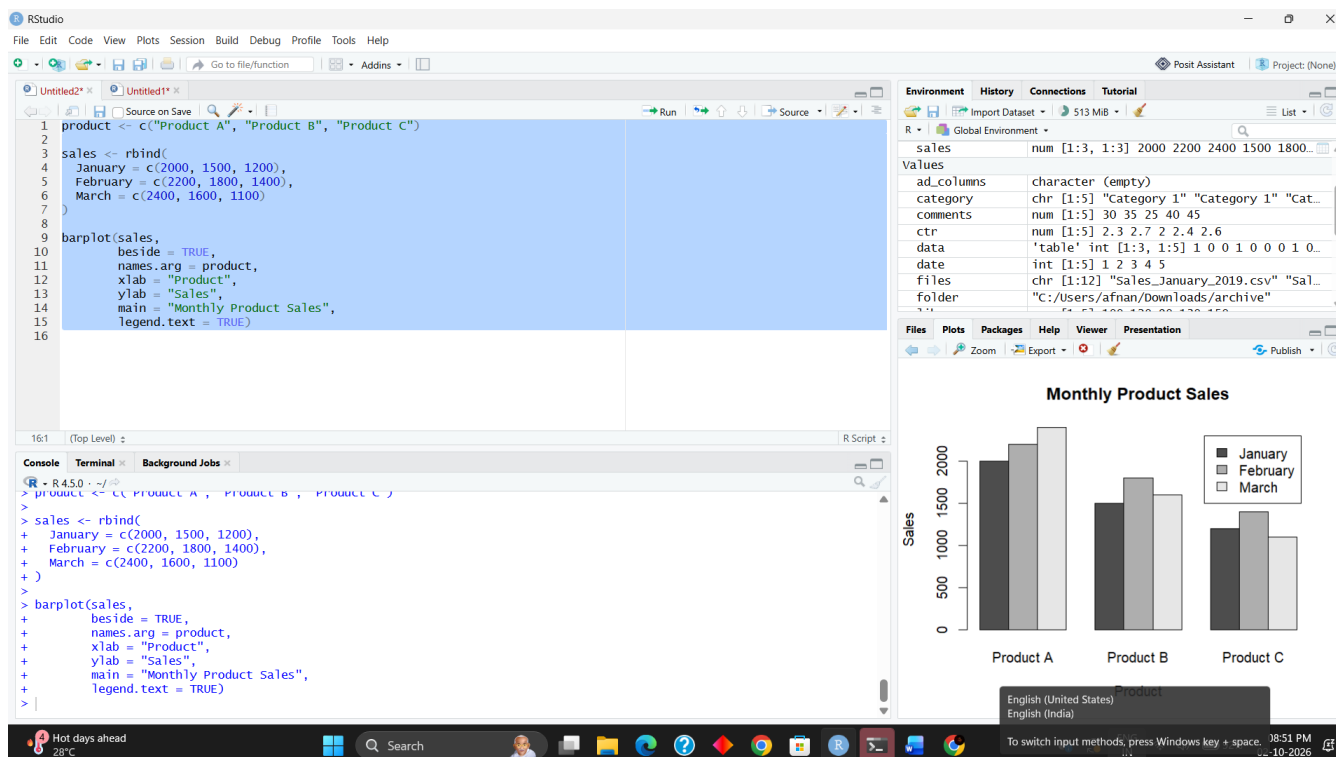
### **PROGRAM**

```
product <- c("Product A", "Product B", "Product C")
```

```
sales <- rbind(  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)
```

```
barplot(sales,  
  beside = TRUE,  
  names.arg = product,  
  xlab = "Product",  
  ylab = "Sales",  
  main = "Monthly Product Sales",  
  legend.text = TRUE)
```

### **OUTPUT:**



## RESULT

The grouped bar chart successfully compares the sales of all products during the first quarter.

## 2. Stacked Area Chart

### AIM

To represent the overall sales trend of all products over three months using a stacked area chart.

### ALGORITHM

1. Create monthly sales vectors for all products.
2. Calculate cumulative sales.
3. Use `polygon()` to create the stacked areas.
4. Add month labels and legend.
5. Display the chart.

### PROGRAM

```
month <- 1:3
```

```
product_A <- c(2000, 2200, 2400)
```

```
product_B <- c(1500, 1800, 1600)
```

```
product_C <- c(1200, 1400, 1100)
```

```
total1 <- product_A
```

```
total2 <- product_A + product_B
```

```
total3 <- product_A + product_B + product_C
```

```
plot(month, total3,
```

```
  type = "n",
```

```
  xaxt = "n",
```

```
  xlab = "Month",
```

```
  ylab = "Sales",
```

```
  main = "Overall Product Sales Trend")
```

```
axis(1, at = 1:3,
```

```
  labels = c("January", "February", "March"))
```

```
polygon(c(month, rev(month)),
```

```
  c(total1, rep(0, 3)))
```

```
polygon(c(month, rev(month)),
```

```
  c(total2, rev(total1)))
```

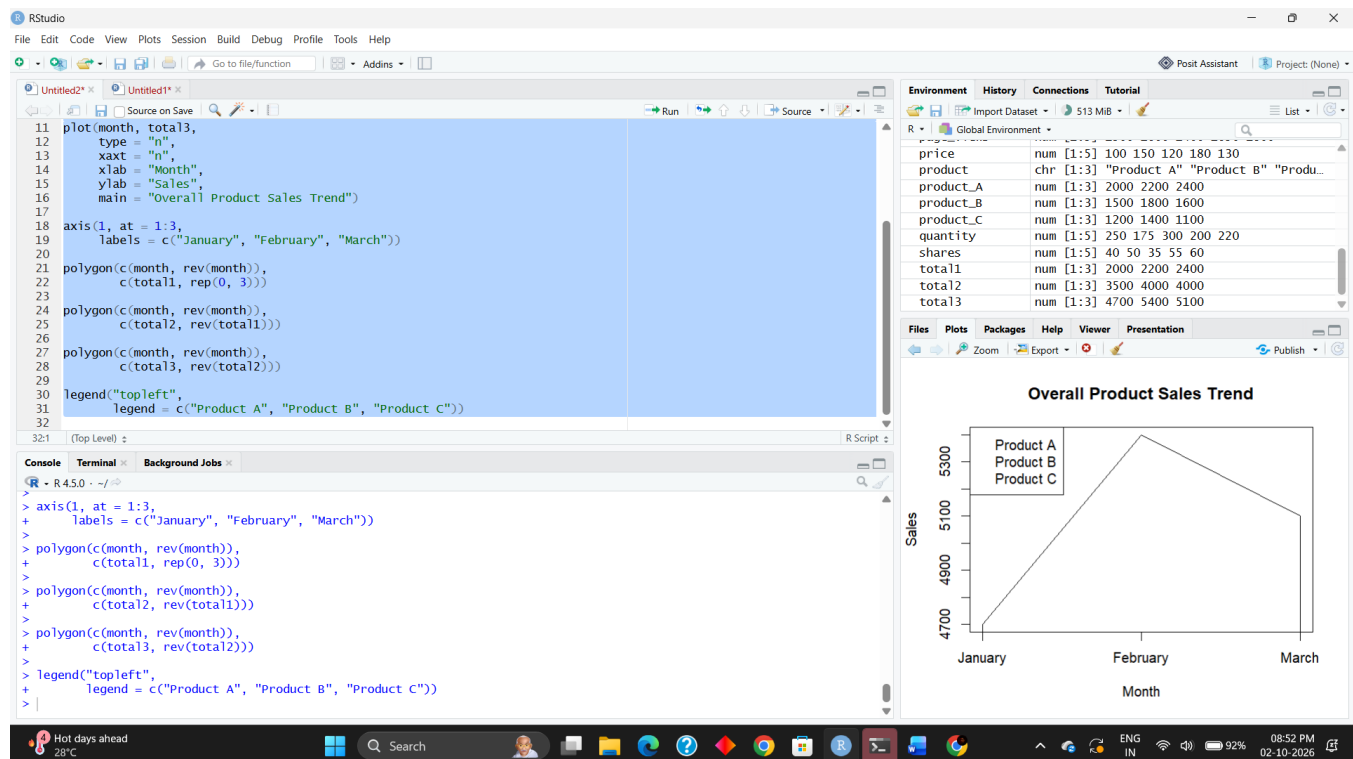
```
polygon(c(month, rev(month)),
```

```
  c(total3, rev(total2)))
```

```
legend("topleft",
```

```
  legend = c("Product A", "Product B", "Product C"))
```

## OUTPUT:



## RESULT

The stacked area chart successfully shows the overall sales trend of the three products over the first quarter.

## 3. Sales Data Table

### AIM

To create a table showing the monthly sales data for each product.

### ALGORITHM

1. Create a data frame containing product names and monthly sales.
2. Store January, February, and March sales.
3. Display the data frame as a table.

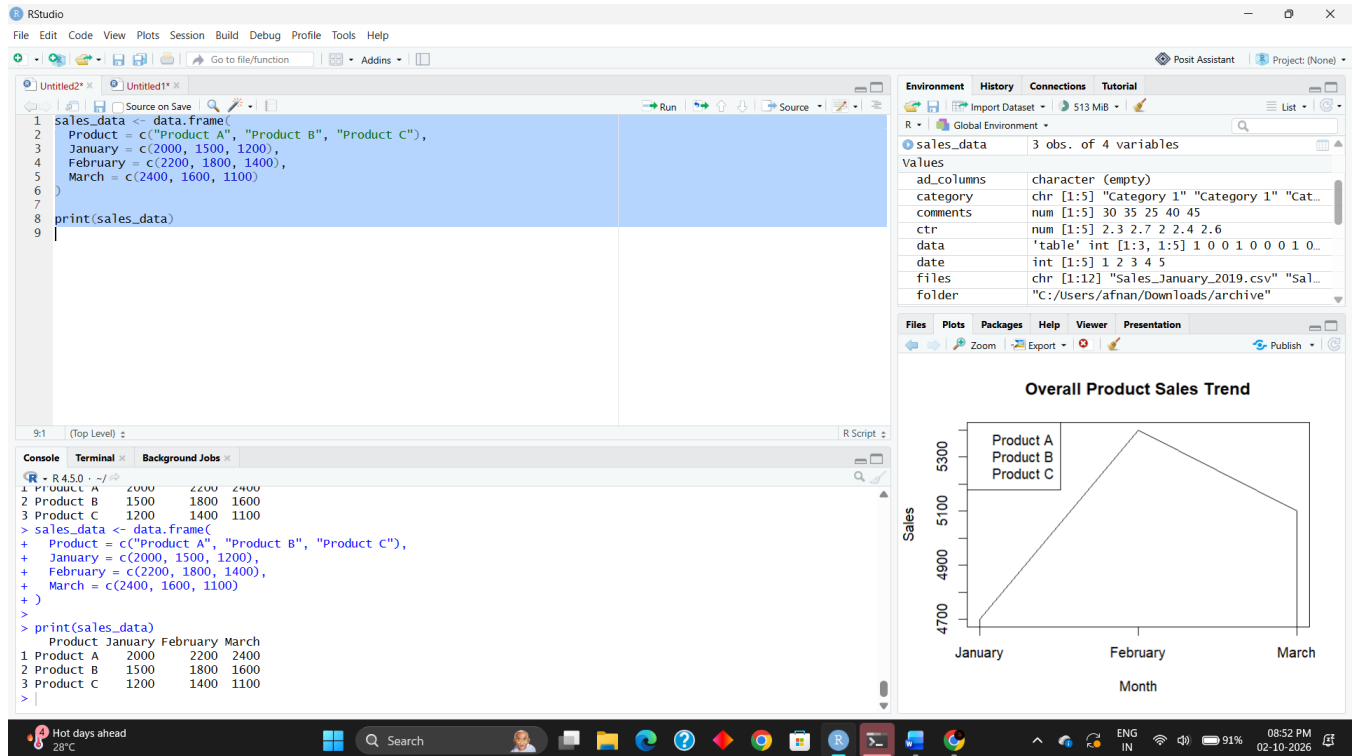
### PROGRAM

```
sales_data <- data.frame(
  Product = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100))
```

)

```
print(sales_data)
```

## OUTPUT:



## RESULT

The table successfully displays the monthly sales data for all three products.

## 4. Tableau Dashboard

### AIM

To develop an interactive Tableau dashboard combining the grouped bar chart, stacked area chart, and sales table.

### ALGORITHM

1. Import the product sales dataset into Tableau.
2. Create a grouped bar chart.
3. Create a stacked area chart.
4. Create a sales data table.
5. Create a new dashboard.
6. Add all three visualizations.

7. Add filters for interactive exploration.

## **PROGRAM**

### **Tableau procedure:**

1. Open Tableau.
2. Import the **Monthly Product Sales** dataset.
3. Create **Grouped Bar Chart**.
4. Create **Stacked Area Chart**.
5. Create **Sales Table**.
6. Select **Dashboard** → **New Dashboard**.
7. Drag all three charts into the dashboard.
8. Add filters and interactive controls.

## **RESULT**

The Tableau dashboard successfully combines the grouped bar chart, stacked area chart, and sales table for interactive analysis of product sales.